Uniform Hölder Nonexpansive Maps on Decomposable Banach Balls

Automatic Math Discovery Run `fa_banach_001`

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Source

Cleon S. Barroso and Valdir Ferreira, *Hölder-contractive mappings, nonlinear extension problem and fixed point free results*, arXiv:2207.03057.

The final open questions ask whether B_X fails the fixed point property for uniformly α-Hölder nonexpansive maps with null minimal displacement when, among other cases, X is isomorphic to a Hilbert space or X is reflexive with an unconditional basis.

5.1 Open questions

We end this paper with the following questions: Let X be an infinite-dimensional Banach space. Does B_X fail the FPP for *uniformly* α-Hölder nonexpansive maps with null minimal displacement if, (1) X is reflexive, (2) X is isomorphic to a Hilbert space or (3) X is reflexive and has an unconditional basis, or then (4) X is the James's space J_2? It should be noted that, intuitively, the general strategy used here to handle ($\mathcal{Q}3$) was to check the failure of the HFPP in a certain subset K of B_X and then try to properly extend it to the whole ball. In some cases this involves the availability of extension results with prescribed conditions, which seems to be the main difficulty in dealing with such a problem.

19

Result

Theorem 1. *Let X be an infinite-dimensional Banach space that splits as*

$$X = Y \oplus Z$$

with Y and Z closed infinite-dimensional subspaces. Then, for every $\alpha \in (0,1)$, there is a map $V : B_X \to B_X$ such that

$$\mathrm{Fix}(V) = \varnothing, \qquad \mathrm{d}(V, B_X) = 0,$$

and

$$\|V^n x - V^n y\| \leq \|x - y\|^\alpha \qquad (x, y \in B_X,\ n \geq 1).$$

Thus V is uniformly α-Hölder nonexpansive.

Corollary 1. *For every $\alpha \in (0,1)$, the source paper's final open questions have affirmative answers in the following cases:*

1. *X is isomorphic to an infinite-dimensional Hilbert space;*
2. *X has an unconditional basis, in particular when X is reflexive and has an unconditional basis.*

Input Lemma From The Run

We use the decomposable-space packet already established in this run: if $X = Y \oplus Z$ with both summands infinite-dimensional, then there is a map $U : B_X \to B_X$ such that

$$\mathrm{Fix}(U) = \varnothing, \qquad \mathrm{d}(U, B_X) = 0, \qquad M := \sup_{n \geq 1} \mathrm{Lip}(U^n) < \infty.$$

This is the result recorded in

`solutions/partial/2307.12958_decomposable_tube_uniformly_lipschitz_q2/`.

The present packet only adds the Hölder-nonexpansive scaling step.

Scaling Lemma

Lemma 1. *Let X be a Banach space. Suppose $U : B_X \to B_X$ satisfies*

$$\mathrm{Fix}(U) = \varnothing, \qquad \mathrm{d}(U, B_X) = 0, \qquad M := \sup_{n \geq 1} \mathrm{Lip}(U^n) < \infty.$$

Then for every $\alpha \in (0,1)$ there is a uniformly α-Hölder nonexpansive map $V : B_X \to B_X$ with $\mathrm{Fix}(V) = \varnothing$ *and* $\mathrm{d}(V, B_X) = 0$.

Proof. Let $C = 2M$ and put

$$\eta = \min\{2, C^{-1/(1-\alpha)}\}.$$

Choose $r > 0$ so small that $2r \leq \eta^\alpha$. In particular $r < 1$. Let $R_r : B_X \to rB_X$ be the radial retraction

$$R_r(x) = \begin{cases} x, & \|x\| \leq r, \\ r\,x/\|x\|, & \|x\| > r. \end{cases}$$

This retraction is 2-Lipschitz on every normed space.

Define the scaled map $S : rB_X \to rB_X$ by

$$S(z) = r\,U(z/r).$$

Then $S^n(z) = r\,U^n(z/r)$, so $\mathrm{Lip}(S^n) \leq M$ for every $n \geq 1$. Moreover, S is fixed-point free and has null minimal displacement on rB_X.

Now set

$$V = SR_r : B_X \to B_X.$$

Since $S(rB_X) \subseteq rB_X$ and R_r is the identity on rB_X,

$$V^n = S^n R_r \qquad (n \geq 1).$$

For $x, y \in B_X$, write $d = \|x - y\|$. Then

$$\|V^n x - V^n y\| \leq \min\{M\|R_r x - R_r y\|, 2r\} \leq \min\{Cd, 2r\}.$$

If $d \leq \eta$, then $Cd \leq d^\alpha$ by the definition of η. If $d \geq \eta$, then $2r \leq \eta^\alpha \leq d^\alpha$. Hence

$$\|V^n x - V^n y\| \leq d^\alpha$$

for all $x, y \in B_X$ and every $n \geq 1$.

There is no fixed point: if $Vx = x$, then $x \in rB_X$, so $R_r x = x$ and $Sx = x$, contradicting $\mathrm{Fix}(S) = \varnothing$.

Finally, since S has null minimal displacement on rB_X, there are $z_j \in rB_X$ such that $\|Sz_j - z_j\| \to 0$. For these points $R_r z_j = z_j$, so $\|Vz_j - z_j\| \to 0$. Thus $\mathrm{d}(V, B_X) = 0$. □

Proof of Theorem 1. Apply the input lemma from the run to obtain U, then apply Lemma 1. □

Proof of Corollary 1. If X is isomorphic to an infinite-dimensional Hilbert space, split the Hilbert space into two closed infinite-dimensional orthogonal summands and transport that decomposition to X through an isomorphism. The transported summands are closed and complemented, so Theorem 1 applies.

If X has an unconditional basis, split the basis into its odd and even subsequences. The associated coordinate subspaces are closed, infinite-dimensional, and complemented by the unconditional coordinate projections. Again Theorem 1 applies. The reflexive unconditional-basis case from the source paper is therefore included. □

Limitations

This does not answer the source paper's arbitrary reflexive case, nor the James-space case. The argument relies on the existence of a decomposable two-infinite-summand structure, because that is what supplies the fixed-point-free uniformly Lipschitzian dynamics with null minimal displacement.

Verification Notes

The only scalar estimate in the new argument is

$$\min\{Cd, 2r\} \leq d^\alpha \qquad (0 \leq d \leq 2),$$

with $\eta = \min\{2, C^{-1/(1-\alpha)}\}$ and $2r \leq \eta^\alpha$. The accompanying script `code/verify_scaling_lemma.py` samples this inequality for representative values of C, r, and α.